UMGS: UAV Multispectral 3D Gaussian Splatting for Spectral-Index Synthesis

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UAV multispectral 3D reconstruction is difficult because multispectral bands are weak geometry drivers: they contain less texture than RGB, and UAV rigs introduce cross-FOV, resolution, and alignment differences. Direct spectral reconstruction can be unreliable, causing per-band geometry/support drift that prevents stable 3D spectral-index synthesis. We introduce UMGS, a two-stage 3D Gaussian Splatting framework that conditions raw multispectral captures with cross-modal rectification and valid-region supervision, reconstructs an RGB Gaussian anchor, and learns multispectral appearance with Gaussian positions, covariances, opacity, and identity frozen. This geometry lock makes spectral products well defined on one shared splat support while retaining band-specific appearance. We instantiate UMGS as independent per-band transfer (UMGS-I) and a joint band-bank branch (UMGS-J). Experiments on a 10-scene UAV-focused evaluation suite show that UMGS preserves shared 3D support while maintaining competitive spectral rendering and index fidelity. Against the available multispectral Gaussian baseline, aligned image-space metrics are scene-dependent, while UMGS provides stronger product readiness and substantially better validation-gated reference-depth consistency.

CCS Concepts: • **Computing methodologies** → **Image-based rendering**; *Reconstruction*; • **Applied computing** → *Earth and atmospheric sciences.*

Additional Key Words and Phrases: multispectral imaging, 3D Gaussian Splatting, 3D reconstruction, UAV

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1 Introduction

UAV multispectral imaging is increasingly used in precision agriculture and environmental monitoring, where decisions are made from cross-band products rather than from a single rendered view [Colomina and Molina 2014; Mathews and Jensen 2013; Turner et al. 2015; Zhang and Kovacs 2012]. A reconstruction used for spectral-index synthesis must therefore satisfy two requirements at once: accurate per-band appearance and stable geometric correspondence across bands. The latter is essential because even small support drift can bias ratio-based measurements such as NDVI, GNDVI, and NDRE [Gitelson et al. 1996; Goetz et al. 1985; Huete 1988; Manolakis et al. 2016; Rouse et al. 1974].

Raw UAV multispectral data make this difficult. Multispectral bands are weak geometry drivers: they often have lower texture and less reliable local features than RGB. Practical UAV rigs add a second source of instability because RGB and multispectral cameras can differ in field of view, resolution, and spectral response. Under raw UAV multispectral captures, direct spectral reconstruction or independent per-band Gaussian optimization can therefore yield models whose geometry and support diverge across bands, preventing a stable unified 3D representation for index synthesis.

Recent radiance-field and Gaussian methods greatly improved training speed and view quality [Barron et al. 2022; Huang et al. 2024; Kerbl et al. 2023; Mildenhall et al. 2020; M"uller et al. 2022; Yu et al. 2024]. Emerging multispectral, hyperspectral, and thermal Gaussian lines [Chen et al. 2024; Lu et al. 2024a; Meyer et al. 2025; Thirgood et al. 2025] further show that spectral rendering and index synthesis are feasible. UMGS addresses the UAV-specific geometry problem: how to exploit RGB as a reliable geometric carrier while learning low-texture spectral appearance on the same 3D support.

We introduce **UMGS**, a UAV multispectral 3D Gaussian Splatting framework for spectral-index synthesis. UMGS formulates multispectral reconstruction as *RGB-anchored geometry locking*: geometry, opacity, and Gaussian identity are learned once from RGB and then frozen during spectral transfer. This design turns RGB into a stable geometric carrier, while each spectral band only learns appearance on the same splat support. We evaluate two implementations of the same constraint:

- **UMGS-I**: independent per-band transfer from a shared RGB anchor.
- **UMGS-J**: joint multispectral training with band-specific appearance banks in one unified checkpoint.

UMGS-I is our conservative default realization, while UMGS-J tests whether the same lock can be implemented as a joint bank representation. Both preserve identical support by construction; they differ only in how spectral appearance parameters are organized and optimized.

For raw UAV captures, the geometry-lock contract also requires a reliable conditioning stage before Gaussian training. Cross-band FOV and resolution mismatch can make naive baselines unreliable, especially for NIR. UMGS therefore conditions raw multispectral inputs with cross-modal rectification, baseline-aware rectification QA, validity masks, and masked spectral supervision. The QA separates strict pass, pass-with-warning, and fail, and logs per-frame matching/stability diagnostics rather than relying on a single opaque gate.

Our main evaluation covers a 10-scene UAV-focused evaluation suite: 7 raw UAV captures and 3 aligned aerial multispectral scenes. The raw track validates end-to-end applicability (alignment, QA, reconstruction, and products). The aligned track removes rectification confounds and supports fair common-mask comparison with MS-Splatting. Across this suite, UMGS-I and UMGS-J are near-tied and consistently preserve shared support. Scratch-MS and Unfrozen-support are useful upper-bound ablations for single-band fitting,

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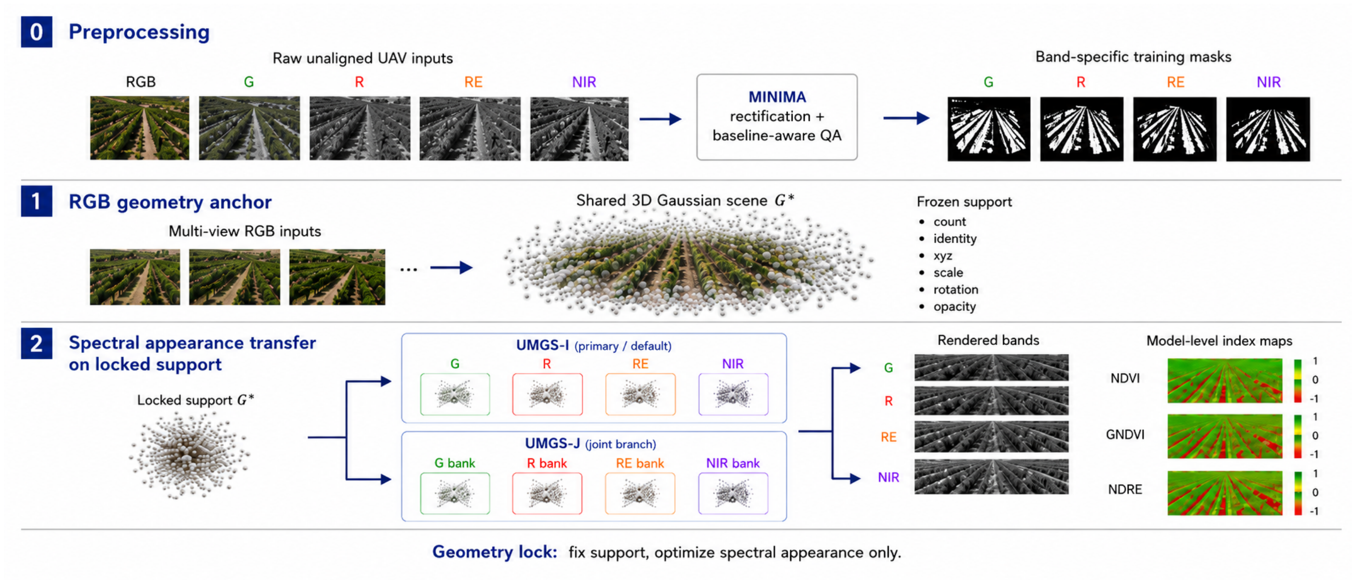

Fig. 1. Overview of UMGS. Raw UAV multispectral captures are challenging because non-RGB bands are weak geometry drivers and direct per-band reconstruction can drift across supports. UMGS first learns an RGB 3D Gaussian anchor, freezes Gaussian support, and then learns spectral appearance either with independent per-band transfer (UMGS-I) or a joint band-bank branch (UMGS-J). Spectral-index products are rendered from the same locked support, and validated diagnostics compare product readiness and reference-depth consistency.

but they do not satisfy the shared-support requirement for stable multispectral index synthesis.

The contributions are:

- A geometry-locked multispectral 3DGS formulation that makes model-level spectral-index synthesis well defined on a shared Gaussian support.
- A raw UAV multispectral conditioning pipeline that combines cross-modal rectification, baseline-aware QA, validity masks, and masked spectral supervision for cross-FOV, cross-resolution captures.
- Two concrete realizations of the same support contract: UMGS-I as the default independent transfer line and UMGS-J as a joint band-bank branch.
- A UAV-focused evaluation suite reporting image quality, spectral indices, cost, common-mask comparison, support-contract audits, and validation-gated reference depth.

2 Related Work

2.1 Radiance Fields and Gaussian Scene Representations

NeRF established neural volumetric rendering as a practical paradigm for novel view synthesis [Mildenhall et al. 2019, 2020]. Follow-up work improved anti-aliasing, unbounded scenes, and training speed, including mip-NeRF/mip-NeRF360, Instant-NGP, Plenoxels, TensoRF, and DVGO [Barron et al. 2022, 2021; Chen et al. 2022; Fridovich-Keil et al. 2022; M"uller et al. 2022; Sun et al. 2022]. Large-scene and explicit-factorized lines such as NSVF, K-Planes, Zip-NeRF, MERF, Mega-NeRF, and Block-NeRF further explored memory/scale trade-offs [Barron et al. 2023; Fridovich-Keil et al. 2023; Liu et al. 2020; Reiser et al. 2023; Tancik et al. 2022; Turki et al. 2022].

3D Gaussian Splatting (3DGS) moved from volumetric integration to explicit anisotropic splats, enabling faster optimization and real-time rendering [Kerbl et al. 2023]. Subsequent work improved alias control, structural priors, compactness, and scene scale, including Mip-Splatting, Scaffold-GS, 2DGS, SuGaR, LightGaussian, Compact3DGS, Dynamic 3DGS, CityGaussian, and CityGaussianV2 [Fan et al. 2024; Guédon and Lepetit 2024; Huang et al. 2024; Lee et al. 2024; Liu et al. 2024, 2025; Lu et al. 2024b; Luiten et al. 2024; Yu et al. 2024]. These methods target rendering quality, memory, or geometry fidelity in RGB-dominant settings. UMGS addresses a different constraint: preserving one stable 3D support while transferring low-texture UAV spectral appearance onto it.

2.2 Multispectral and Thermal Gaussian Reconstruction

Multispectral remote sensing has long emphasized calibration, cross-band consistency, and physically meaningful indices [Gitelson et al. 1996; Goetz et al. 1985; Huete 1988; Manolakis et al. 2016; Rouse et al. 1974]. In UAV workflows, geometric preprocessing and modality alignment are often the limiting factor for downstream reliability [Colomina and Molina 2014; Mathews and Jensen 2013; Turner et al. 2015; Zhang and Kovacs 2012]. While deep learning for spectral imagery is mature in classification/segmentation tasks [Audebert et al. 2019; Li et al. 2019; Paoletti et al. 2019; Zhong et al. 2018], 3D multispectral scene reconstruction is still emerging.

Recent Gaussian-based spectral works include MS-Splatting for multi-camera multispectral NVS, HyperGS for hyperspectral latent rendering, MSGS for wavelength-aware Gaussian modeling, and thermal-domain lines such as ThermalGaussian and Thermal3D-GS [Chen et al. 2024; Lu et al. 2024a; Meyer et al. 2025; Thirgood et al. 2025; Zhang et al. 2026]. These works confirm the feasibility of spectral Gaussian rendering and, in several cases, already store multiple spectral or modal signals in one Gaussian-based representation. MS-Splatting, for example, learns compact per-splat features and decodes spectral values with an MLP; HyperGS performs hyperspectral rendering in a learned latent space; ThermalGaussian learns registered RGB/thermal Gaussians after camera calibration. Our focus is complementary: raw UAV multispectral deployment, where spectral bands are weak geometry drivers and RGB/MS cameras can differ in FOV, resolution, and alignment. Rather than relying on each spectral band to define its own geometry, UMGS uses RGB as the geometric carrier and learns spectral appearance on the locked support.

2.3 SfM, Matching, and Cross-Modal Alignment

Practical reconstruction still depends on robust SfM/MVS foundations [Fischler and Bolles 1981; Hartley and Zisserman 2004; Sch"onberger and Frahm 2016; Sch"onberger et al. 2016]. Handcrafted and learned feature matching both remain relevant, from SIFT/SURF/ORB and ECC-style alignment to SuperPoint, SuperGlue, LoFTR, LightGlue, and RoMa [Bay et al. 2008; DeTone et al. 2018; Edstedt et al. 2024; Evangelidis and Psarakis 2008; Lindenberger et al. 2023; Lowe 2004; Rublee et al. 2011; Sarlin et al. 2020; Sun et al. 2021]. Robust estimators (LO-RANSAC, USAC, MAGSAC) are essential under modality gaps and outlier-heavy correspondences [Barath et al. 2019; Chum et al. 2003; Raghunandan et al. 2013].

Our raw-scene conditioning follows this systems view: it reports match quality and geometric stability separately, preserves legacy strict criteria for strong baselines, uses a selective warning path only for weak-baseline/high-modality-gap cases, and carries rectification validity into masked spectral supervision. It is a practical raw-capture component rather than a new matcher.

2.4 Depth and Geometry Diagnostics

Depth-based reconstruction metrics are widely used in multi-view stereo and neural reconstruction [Gu et al. 2020; Wang et al. 2021; Yao et al. 2018]. However, raw UAV multispectral captures usually lack globally calibrated metric scale, and dense MVS references can fail under low texture or repeated agricultural structure. We therefore treat reference-depth as a validation-gated diagnostic rather than an unconditional accuracy benchmark: only references with valid geometric depth, normals, fusion, and support are eligible for quantitative reporting. This keeps our geometry discussion closer to repeatability/self-consistency under controlled probes than to absolute-meter accuracy claims.

3 Method

3.1 Geometry-Locked Multispectral 3DGS

Given RGB and multispectral UAV captures, our goal is to synthesize spectral novel views and spectral indices on one consistent 3D support. We write a Gaussian model as a support–appearance pair

$$M_b = (\mathcal{S}, \mathcal{A}_b),$$

where $\mathcal{S} = \{N, \mathbf{x}_i, \mathbf{s}_i, \mathbf{q}_i, \alpha_i\}_{i=1}^{N}$ contains Gaussian count, identity, position, scale, rotation, and opacity, and $\mathcal{A}_b$ contains band-specific appearance for band b. A spectral index is a model-level product only if all participating bands are rendered from the same $\mathcal{S}$. Otherwise, an index becomes a post-hoc ratio between unrelated per-band reconstructions.

UMGS therefore separates support estimation from spectral appearance learning. Stage 1 estimates a single RGB support:

$$(\mathcal{S}^\star, \mathcal{A}^\star_{RGB}) = \arg\min_{\mathcal{S}, \mathcal{A}_{RGB}} \sum_{v} \mathcal{L}\big(R(\mathcal{S}, \mathcal{A}_{RGB}; v), I^{v}_{RGB}\big),$$

using the standard 3DGS photometric training pipeline [Kerbl et al. 2023]. Stage 2 freezes $\mathcal{S}^\star$ and optimizes only spectral appearance:

$$\mathcal{A}^\star_b = \arg\min_{\mathcal{A}_b} \sum_{v} \mathcal{L}\big(R(\mathcal{S}^\star, \mathcal{A}_b; v), I^{v}_{b}\big), \quad b \in \{G, R, RE, NIR\}.$$

This is the geometry-lock contract. It fixes Gaussian identity, position, covariance parameters, and opacity during spectral training, so the i-th Gaussian denotes the same support element for every band.

3.2 Spectral-Index Synthesis on Shared Support

With locked support, each rendered spectral image is a view of the same Gaussian topology. Index synthesis is then defined directly over consistent band predictions, for example:

$$NDVI(v, \mathbf{u}) = \frac{R(\mathcal{S}^\star, \mathcal{A}_{NIR}; v, \mathbf{u}) - R(\mathcal{S}^\star, \mathcal{A}_{R}; v, \mathbf{u})}{R(\mathcal{S}^\star, \mathcal{A}_{NIR}; v, \mathbf{u}) + R(\mathcal{S}^\star, \mathcal{A}_{R}; v, \mathbf{u}) + \epsilon}.$$

The same support is used for GNDVI and NDRE. This is the core distinction from direct per-band reconstruction: per-band PSNR can improve while the support contract is broken, in which case model-level products are no longer valid in our protocol.

3.3 UMGS-I and UMGS-J

UMGS-I (default independent transfer). UMGS-I initializes each spectral band from the RGB anchor, freezes $\mathcal{S}^\star$, and trains one appearance model $\mathcal{A}_b$ per band. It is the conservative default because it exports standard 3DGS-compatible per-band artifacts and keeps the training path simple.

UMGS-J (joint band-bank branch). UMGS-J keeps the same frozen $\mathcal{S}^\star$ but stores band-specific appearance banks in one joint checkpoint. At each training step, one active bank is optimized while the shared support and inactive banks remain unchanged. UMGS-J tests whether the geometry-lock principle is tied to independent transfer or also holds under a joint multispectral parameterization.

Support-contract audit. For every product-ready output we audit:

- identical Gaussian count and identity across bands;
- identical xyz, scale, rotation, and opacity;
- appearance-only updates in Stage 2;
- successful model-level product construction.

Scratch-MS and Unfrozen-support are therefore diagnostic ablations. They can render individual bands, but if their count or support parameters differ across bands, they are not valid model-level spectral product models.

3.4 Raw UAV Multispectral Conditioning

Raw UAV scenes require spectral-to-RGB conditioning before reconstruction. We estimate per-band homographies using deterministic candidate sampling and robust aggregation from cross-modal feature matches. Each accepted homography defines both a rectified spectral image and a validity mask induced by the warp footprint. During spectral training, photometric supervision is evaluated only on valid pixels, so black borders, undefined warp regions, and non-overlapping cross-FOV areas do not bias the appearance model.

Since naive scaling baselines can be unreliable under cross-FOV and modality gaps, we use a baseline-aware QA protocol with strict pass, pass-with-warning, and fail states. The warning path is restricted to weak-baseline/high-modality cases that satisfy match quality, homography stability, gradient gain, edge-floor, and outlier-ratio checks. This conditioning stage makes raw multi-camera UAV captures usable for the geometry-lock contract; aligned inputs use the same UMGS representation without this rectification stage.

3.5 Validation-Gated Reference Depth

When a held-out reference source passes visual, support, and camera-frame validation, we diagnose relative depth agreement on probe views:

$$\delta_{rel} = \frac{D_{model} - D_{ref}}{D_{ref}}.$$

Because raw UAV reconstructions are not globally metric, we do not claim dense absolute geometry accuracy. Failed or frame-misaligned reference sources are excluded rather than forced into averages. The resulting depth table is a validation-gated consistency diagnostic, complementary to the shared-support audit.

4 Experiments

4.1 Setup

Scenes. We evaluate the main paper on a 10-scene UAV-focused evaluation suite: 7 raw UAV scenes (UMGS-Field, Vineyard-01–Vineyard-06) and 3 aligned aerial multispectral scenes (Aerial-Golf, Aerial-Lake, Aerial-Solar). Raw scenes test rectification, QA, reconstruction, and products. Aligned scenes remove rectification confounds and are used for external common-mask comparison. Four additional non-UAV official aligned scenes are reported only in the supplement and are not part of the main 10-scene evaluation suite.

Methods. We report UMGS-I as the default realization and UMGS-J as a joint band-bank branch of the same geometry-lock contract. MS-Splatting is the external baseline on aligned aerial scenes. Scratch-MS and Unfrozen-support are representative single-band upper-bound ablations; they are not product models because they do not preserve the shared Gaussian support required by model-level multispectral fusion.

Metrics. Image quality: PSNR/SSIM/LPIPS. Spectral means average G/R/RE/NIR. Index fidelity: MAE/RMSE/PSNR/SSIM for NDVI, GNDVI, NDRE. Cost uses non-invasive GPU traces and exported Gaussian counts. Aligned external comparison uses a common-mask protocol; † marks UMGS renders resized to the external adapter resolution for comparison only, not distinct trained variants. We report geometry only as a validation-gated reference-depth diagnostic. Scenes whose reference mesh or camera-frame alignment failed inspection are excluded from depth tables rather than forced into an average.

4.2 Main Results on UAV Scenes

The main question is whether locking RGB geometry sacrifices spectral rendering quality. Table 1 shows that it does not: UMGS-I and UMGS-J remain near-identical on aggregate image and index quality. On the 10-scene UAV-focused evaluation suite, both reach 0.843 spectral SSIM and 0.844 index SSIM; UMGS-I has a small edge on index RMSE (0.075 vs. 0.077). We therefore use UMGS-I as the conservative primary line and treat UMGS-J as a joint-parameterization branch with comparable performance.

4.3 Index-Focused Analysis

Spectral-index synthesis is the downstream use case that motivates shared support. Table 2 breaks out NDVI, GNDVI, and NDRE. UMGS-I and UMGS-J are close on NDVI and GNDVI, while NDRE shows a larger RMSE/PSNR gap favoring UMGS-I. The joint-bank design is therefore viable but not consistently superior.

4.4 Reference-Depth Diagnostics

Table 3 reports relative-depth consistency only for reference sources that passed visual and protocol validation. The diagnostic is not an absolute geometry benchmark: it evaluates agreement with a scene-specific reference under held-out probe cameras. On the four validated scenes, UMGS-I and UMGS-J have lower relative depth error and higher agreement rates than the retained MS-Splatting adapter (AbsRel 0.023 vs. 0.195, Ag@5% 0.867 vs. 0.526). Band-only has a low mean AbsRel only on the small fraction where it yields valid depth (Valid 0.410), so it should not be read as a better geometry model. Unfrozen-support can be close in depth consistency, but it fails the shared-support/product contract and is therefore not a valid spectral product model. Figure 2 shows the same diagnostic on our project-owned UMGS-Field capture. Scene-level depth breakdowns and all-band visual grids are provided in the supplement.

4.5 Aligned External Comparison

Aligned scenes remove the raw-rectification factor and isolate the reconstruction model. Under this common-mask setting, Table 4 shows competitive, scene-dependent image-space behavior rather than universal dominance: UMGS has lower LPIPS, matches mean SSIM/SAM, and remains within a small margin in PSNR and spectral RMSE. Scene-level results in Table 5 are also mixed. We therefore use this block to establish aligned competitiveness, while the stronger distinction appears in product readiness, shared-support validity, and validation-gated reference-depth consistency.

Raw unaligned data test a different question: whether an external method can be applied under its retained-subset path. We keep this retained-subset evidence in the supplement as an applicability diagnostic rather than a main raw comparison, because it does not represent the full raw protocol used by UMGS-I/UMGS-J.

Table 1. Main UMGS results over the 10-scene UAV-focused evaluation suite. RGB metrics for UMGS-J use the shared UMGS-I RGB anchor. Spectral metrics average G/R/RE/NIR. Index metrics average NDVI/GNDVI/NDRE. Geometry diagnostics require validated reference sources and are therefore kept separate from this main quality table.

Method	Split	RGB PSNR	RGB SSIM	MS PSNR	MS SSIM	MS LPIPS	Idx. RMSE	Idx. SSIM
UMGS-I	Raw 7	24.39	0.840	27.40	0.850	0.136	0.065	0.869
UMGS-I	Aligned 3	25.48	0.810	26.50	0.828	0.230	0.098	0.787
UMGS-I	UAV 10	24.72	0.831	27.13	0.843	0.164	0.075	0.844
UMGS-J	Raw 7	24.39	0.840	27.37	0.850	0.136	0.067	0.869
UMGS-J	Aligned 3	25.48	0.810	26.50	0.828	0.230	0.098	0.787
UMGS-J	UAV 10	24.72	0.831	27.11	0.843	0.164	0.077	0.844

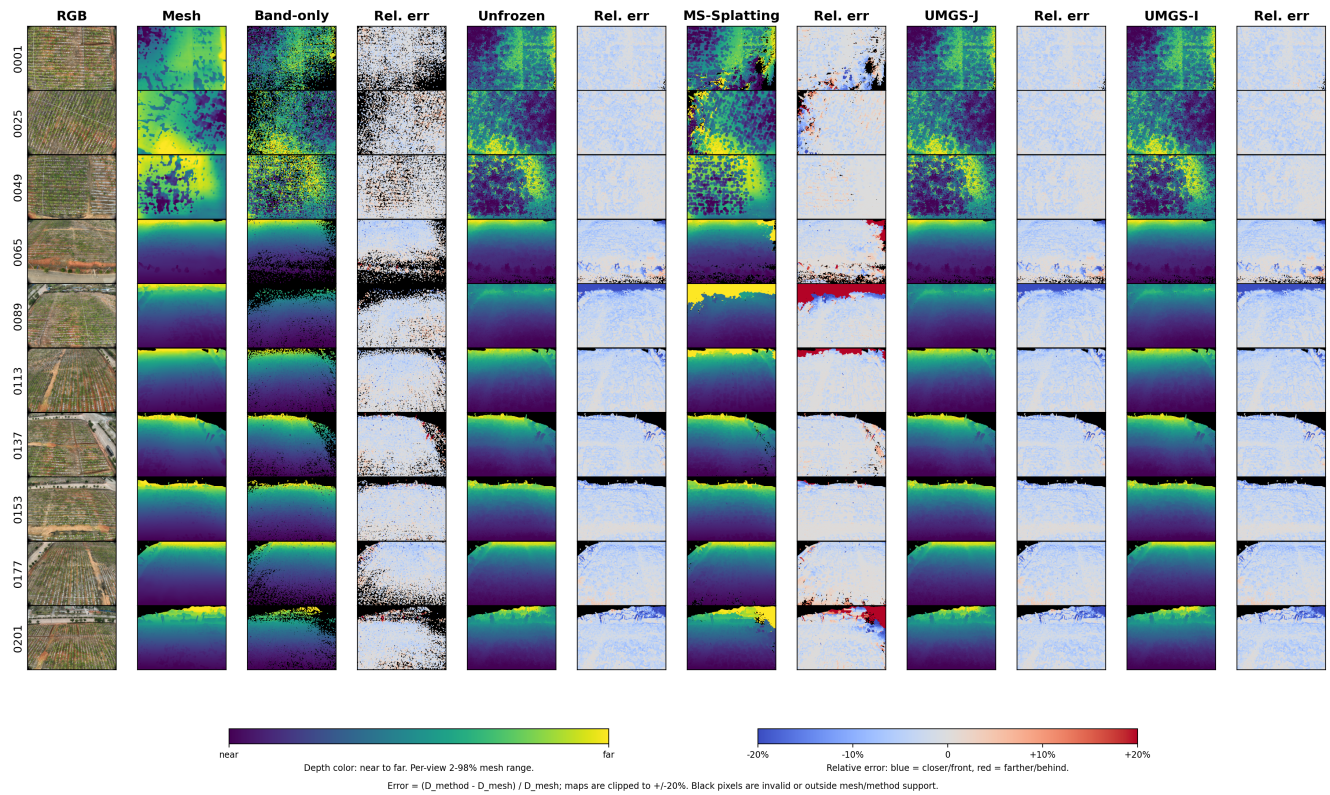

Fig. 2. NIR-band reference-depth diagnostic on UMGS-Field, our project-owned UAV capture. Rows are held-out views. Columns show the input image, mesh reference depth, and depth/error pairs for Band-only, Unfrozen-support, MS-Splatting, UMGS-J, and UMGS-I. UMGS-I and UMGS-J preserve broad valid support with low relative error, while the retained external adapter exhibits larger local deviations.

Table 2. UAV-focused 10-scene spectral-index fidelity by vegetation index.

Method	Index	MAE	RMSE	PSNR	SSIM
UMGS-I	NDVI	0.058	0.083	28.49	0.847
UMGS-I	GNDVI	0.059	0.083	28.65	0.823
UMGS-I	NDRE	0.041	0.059	31.53	0.864
UMGS-J	NDVI	0.058	0.083	28.49	0.847
UMGS-J	GNDVI	0.059	0.083	28.66	0.823
UMGS-J	NDRE	0.043	0.063	31.31	0.863

4.6 Shared-Support Contract Ablation

This ablation tests whether geometry locking is merely conservative or actually necessary. Table 6 compares the locked UMGS variants with two upper-bound ablations on UMGS-Field, Vineyard-02, and Aerial-Lake. This is a separate support-contract diagnostic subset and is not averaged with the four-scene reference-depth diagnostic. The unfrozen branch improves selected image metrics, but it is not a valid product model: on Aerial-Lake, product construction fails the shared-support audit with an xyz mismatch of 6.84×10^{-2}. Scratch-MS diverges more strongly, including Gaussian-count mismatch across bands. These failures occur when multispectral products are

Table 3. Validation-gated reference-depth consistency averaged over four representative scenes (UMGS-Field, Vineyard-01, Vineyard-03, and Aerial-Golf). MS-Splatting is evaluated through its retained-subset adapter where learned Gaussian depth can be exported. Scratch-MS and Unfrozen-support are diagnostic ablations rather than product-ready multispectral models; low error under low valid support should not be interpreted as better geometry.

Method / scope	Valid ↑	AbsRel ↓	Ag@1% ↑	Ag@5% ↑	Ag@10% ↑
UMGS-I	0.951	0.023	0.465	0.867	0.922
UMGS-J	0.951	0.023	0.465	0.867	0.922
Scratch-MS	0.410	0.020	0.211	0.377	0.399
Unfrozen-support	0.958	0.023	0.459	0.871	0.929
MS-Splatting retained	0.801	0.195	0.348	0.526	0.610

Table 4. Aligned aerial common-mask comparison against MS-Splatting. Lower is better for LPIPS, 4-band RMSE, and SAM. Daggered UMGS rows are resolution-aligned comparison views, not separate training variants.

Method	PSNR	SSIM	LPIPS	4-band RMSE	SAM°
UMGS-I†	27.01	0.853	0.215	0.0545	5.38
UMGS-J†	27.01	0.853	0.215	0.0545	5.38
MS-Splatting	27.04	0.853	0.244	0.0541	5.38

assembled, showing that per-band quality gains alone are insufficient if support consistency is lost.

4.7 Cost

Finally, Table 7 reports trace-based cost indicators from AutoDL runs. UMGS-J records higher peak and average VRAM because it maintains a joint multispectral training state, while both UMGS variants keep the same exported Gaussian count by construction. These are non-invasive run traces, not a standardized FPS microbenchmark.

5 Limitations

First, aligned external comparison is scene-dependent in per-scene image metrics. UMGS is competitive on the aligned common-mask protocol and lower in LPIPS on the UAV aligned subset, but individual scenes can favor MS-Splatting in PSNR or spectral RMSE. We therefore emphasize protocol reliability, product readiness, shared-support validity, and reference-depth behavior rather than claiming that every scalar metric wins on every scene.

Second, UMGS-I and UMGS-J are very close numerically. This suggests that the geometry-lock principle matters more than the specific independent or joint spectral parameterization. In the current paper, UMGS-J is best interpreted as a strong alternative branch rather than a strict replacement for UMGS-I.

Third, our geometry diagnostics are validation-gated reference-depth checks, not dense metric ground truth. We report only scenes whose reference mesh and camera frame pass validation, and we exclude collapsed dense-MVS or misaligned reference attempts from quantitative tables. This conservative policy avoids reporting invalid reference geometry, but it also means the current paper does not replace LiDAR-grade absolute geometry diagnostics.

Fourth, cost reporting is intentionally non-invasive. The reported VRAM/utilization/power/size statistics are traceable AutoDL run records, but a dedicated runtime microbenchmark (e.g., standardized per-view FPS across methods and resolutions) is still future work.

6 Conclusion

We presented UMGS, a geometry-locked multispectral 3D Gaussian Splatting framework for UAV spectral-index synthesis. UMGS anchors geometry in RGB and learns spectral appearance on fixed support, so cross-band products are formed on consistent Gaussian topology rather than loosely coupled per-band reconstructions.

The experiments support four conclusions. First, RGB-anchored geometry locking is the central design choice: it preserves support consistency even when weaker spectral bands or unfrozen per-band optimization can improve isolated image metrics. Second, raw UAV multispectral conditioning is necessary in practice, because cross-FOV and cross-resolution captures require rectification, QA, and valid-region supervision before spectral transfer. Third, the same geometry-lock principle remains stable under two parameterizations. UMGS-I is the conservative primary line, while UMGS-J provides a viable joint branch with comparable behavior. Fourth, against the available multispectral Gaussian baseline, aligned image metrics are competitive and scene-dependent, while UMGS provides stronger product readiness and better validation-gated reference-depth consistency by the primary relative-error diagnostics.

The broader implication is that UAV multispectral reconstruction should treat shared 3D support as a first-order constraint for spectral-index synthesis, not as a secondary check after single-band rendering optimization.

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Table 5. Scene-level aligned aerial common-mask comparison for UMGS-I† and MS-Splatting. PSNR/SSIM/LPIPS average G/R/RE/NIR; 4-band RMSE and SAM evaluate the spectral vector.

Scene	UMGS PSNR	MMS PSNR	UMGS SSIM	MMS SSIM	UMGS LPIPS	MMS LPIPS	UMGS RMSE	MMS RMSE	UMGS SAM	MMS SAM
Aerial-Golf	31.33	32.12	0.928	0.935	0.176	0.234	0.0300	0.0273	2.77	3.11
Aerial-Lake	22.02	22.79	0.740	0.731	0.350	0.374	0.0887	0.0817	8.84	7.88
Aerial-Solar	27.72	26.21	0.893	0.893	0.117	0.123	0.0448	0.0532	4.52	5.16

Table 6. Shared-support contract ablation on three representative scenes (UMGS-Field, Vineyard-02, and Aerial-Lake). Spectral metrics average G/R/RE/NIR; index metrics average NDVI/GNDVI/NDRE. Scratch-MS and Unfrozen-support are single-band upper-bound ablations, not valid product models, because their shared-support audits fail.

Method	MS PSNR	MS SSIM	MS LPIPS	Idx. RMSE	Shared	Products
UMGS-I	25.35	0.786	0.228	0.100	yes	yes
UMGS-J	25.33	0.786	0.228	0.100	yes	yes
Scratch-MS	25.88	0.780	0.289	0.116	no	invalid
Unfrozen-support	26.27	0.794	0.218	0.105	no	invalid

Table 7. Trace-based cost indicators from AutoDL runs. VRAM, utilization, and power are sample-weighted over non-invasive GPU traces; Gaussian counts are exported PLY vertex counts for evaluation models.

Method	Peak GB	Avg. GB	Util. %	Power W	Gauss. M
UMGS-I	27.45	12.09	61.4	226.4	4.63
UMGS-J	35.04	18.70	77.0	274.9	4.63

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